

E1

Consider the circuit in Fig. 1. The circuit parameters at $T = 25^\circ\text{C}$ are the following:

$V_{CC} = 8\text{ V}$; $R_{E1} = 0.68\text{ k}\Omega$; $R_{E2} = 4.7\text{ k}\Omega$; $R_P = 6.8\text{ k}\Omega$; $R_S = 10\text{ k}\Omega$; $C_S = 1\text{ }\mu\text{F}$.

$Q_{1,3,4}$ (PNP BJT): $V_{BEp} = -0.7\text{ V}$; $\beta_{Fp} = \beta_{Op} = 150$; $V_{CEsatp} = -0.2\text{ V}$; $r_{Op} = 280\text{ k}\Omega$;

Q_2 (NPN BJT): $V_{BE n} = 0.7\text{ V}$; $\beta_{Fn} = \beta_{On} = 200$; $V_{CEsatn} = 0.2\text{ V}$; $r_{On} = +\infty$;

$V_T = 26\text{ mV}$.

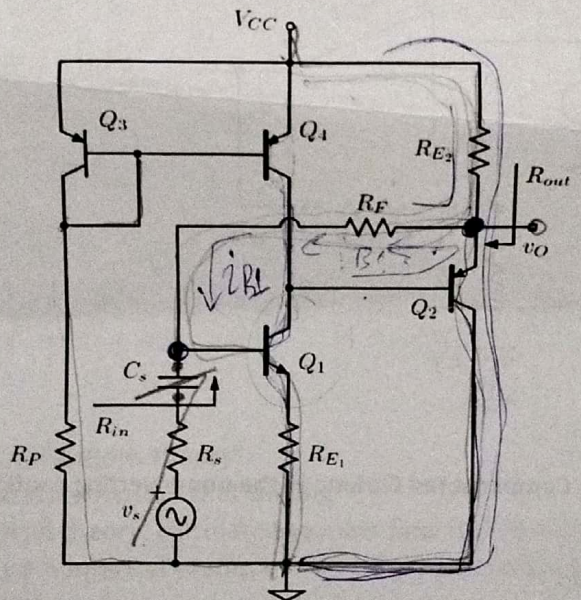


Fig. 1 – A case of two-stage feedback amplifier with active bias.

Considering *all* capacitors to be equivalent to *open circuits*, the current mirror transistors Q_3 and Q_4 to be *identical* and operating at the *same temperature* and neglecting *Early effect* ($\rightarrow I_{C4} = I_{C3}$), determine:

1. the value of R_F that ensures $V_O = -V_{CE2}$ is biased at 4 V ;
2. the operating points (V_{CE} , I_C) of Q_1 and Q_2 .

Assuming that, at the frequencies of interest (i.e. at mid-band), capacitor C_S is equivalent to a *short circuit* and taking into account the current mirror loading effect on Q_1 , determine also:

3. the voltage gain $A_v = v_o/v_s$ of the amplifier;
4. the input resistance R_{in} indicated in the figure;

5. the output resistance R_{out} indicated in the figure.

Finally, considering the effect of capacitor C_s :

6. the low frequency bandwidth limit f_l of the amplifier.

E2

Consider the operational amplifier configuration shown in Fig. 2. The circuit parameters are the following:

$R_1 = 1 \text{ k}\Omega$; $R_2 = 68 \text{ k}\Omega$; $C_1 = 1 \text{ nF}$.

OpAmp: $A_{d0} = 10^5 \text{ [V/V]}$, $\omega_0 = 10 \text{ rad/s}$.

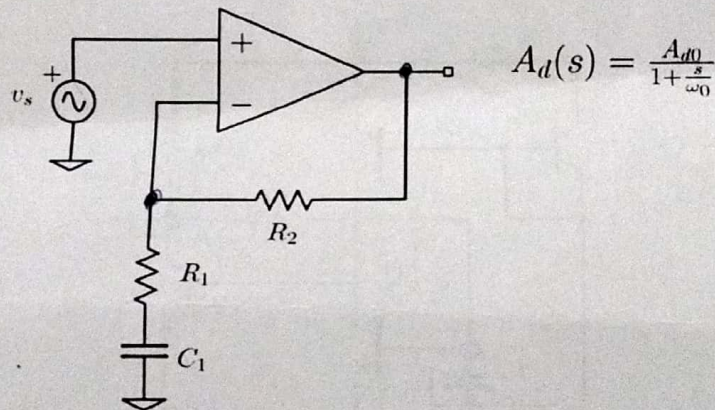


Fig. 2 – Compensated OpAmp in the non-inverting configuration.

Determine:

1. the expression of the *ideal* circuit voltage gain $A_v(s) = v_o/v_s$;
2. the amplifier bandwidth and *phase* margin.

Then, determine also:

3. the value of capacitor C_1 that makes the phase margin of the circuit equal to 60° ;
4. the amplifier bandwidth with the modified C_1 value.